

EARTHQUAKE HAZARD AND VULNERABILITY PROFILE OF PUNJAB AND CHANDIGARH



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Master's in Disaster Management

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INTRODUCTION

Earthquakes are among the most destructive natural hazards, causing widespread loss of life, damage to infrastructure, environmental degradation, and significant economic losses. They occur due to the sudden release of energy within the Earth's crust, resulting in seismic waves that shake the ground. Most earthquakes are caused by the movement of tectonic plates along geological faults, although some may also be induced by human activities such as mining, reservoir impoundment, and underground explosions. The impact of an earthquake depends not only on its magnitude but also on factors such as depth, distance from the epicenter, population density, quality of construction, and the level of disaster preparedness.

The Earth's lithosphere is divided into several tectonic plates that are constantly moving. India lies on the **Indian Plate**, which is moving northward and colliding with the **Eurasian Plate** at a rate of about 5 cm per year. This ongoing collision has formed the Himalayan mountain range and continues to generate significant tectonic stress, making the Himalayan region one of the most seismically active areas in the world. As a result, India experiences frequent earthquakes of varying intensity, particularly in the northern and northeastern regions.

According to the **Bureau of Indian Standards (BIS)** seismic zoning map (IS 1893: Part 1), India is divided into four seismic zones—Zone II, Zone III, Zone IV, and Zone V—representing areas of low to very high earthquake risk. Approximately 59% of the country's land area is vulnerable to moderate to severe earthquakes. To reduce seismic risk, the **National Disaster Management Authority (NDMA)** has issued guidelines promoting earthquake-resistant construction, retrofitting of vulnerable structures, enforcement of seismic building codes, public awareness, and capacity building. The **National Centre for Seismology (NCS)** continuously monitors seismic activity across the country and provides earthquake information for disaster management and research.

Although **Punjab** is not located directly within the Himalayan seismic belt, it remains vulnerable because of its proximity to the active Himalayan fault systems, including the Himalayan Frontal Thrust and the Main Boundary Thrust. According to the BIS seismic zoning map, Punjab falls within **Seismic Zones II, III, and IV**, while **Chandigarh** is located in **Seismic Zone IV**, indicating a relatively high level of earthquake hazard. In addition to natural factors, rapid urbanization, increasing population density, and non-engineered construction practices increase the region's earthquake risk.

This report focuses on the earthquake hazard and vulnerability of **Punjab**, while considering the broader tectonic and seismic setting of India. It examines the geological background, seismic zonation, earthquake history, vulnerability profile, disaster preparedness, institutional framework, and mitigation strategies to provide a comprehensive understanding of earthquake risk and resilience in Punjab.

Chapter 1: Fundamentals of Earthquakes

1.1 Introduction to Earthquakes

An earthquake is the sudden shaking of the Earth's surface caused by the rapid release of energy stored within the Earth's crust. This energy is released when rocks break or slip along geological faults due to tectonic forces, generating seismic waves that travel through the Earth. Earthquakes vary in magnitude and intensity and can range from minor tremors to catastrophic events causing extensive damage to life, property, and the environment.

Most earthquakes occur along tectonic plate boundaries where the Earth's lithospheric plates interact. However, earthquakes can also occur within plates (intraplate earthquakes) due to the reactivation of ancient faults or may be induced by human activities such as mining, reservoir impoundment, and underground explosions.

1.2 Causes of Earthquakes

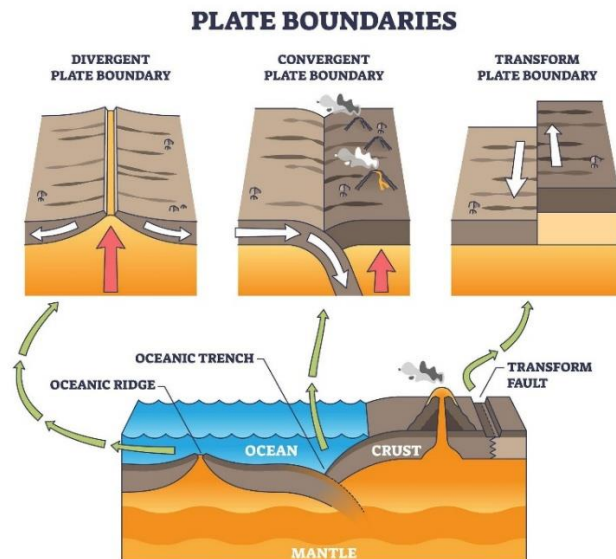
Earthquakes are caused by both natural and human-induced processes. The major causes include:

- Movement of tectonic plates.
- Sudden slip along geological faults.
- Volcanic activity.
- Collapse of underground caves or mines.
- Reservoir-induced seismicity.
- Human activities such as mining, drilling, and underground explosions.

The majority of destructive earthquakes are tectonic in origin and occur due to stress accumulation and sudden release along active faults.

1.3 Plate Tectonics and Earthquake Generation

This diagram shows the three main types of plate boundaries that cause earthquakes:



Divergent Boundary (Plates Move Apart)

- Two tectonic plates move away from each other.
- Magma rises and creates new crust.
- Example: Mid-ocean ridges.

Convergent Boundary (Plates Collide)

- Two plates move toward each other.
- One plate may slide beneath the other (subduction).
- Causes powerful earthquakes, volcanoes, and mountain building.

Transform Boundary (Plates Slide Past)

- Plates slide horizontally past each other.
- Friction builds up and is released as earthquakes.
- Example: San Andreas Fault.

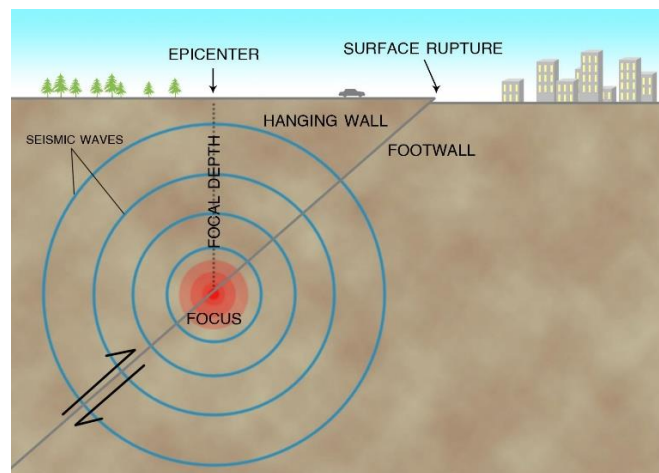
1.4 Faults and Types of Faults

A fault is a fracture or crack in the Earth's crust along which rocks have moved due to tectonic forces.

The major types of faults are:

- **Normal Fault:** Occurs due to tensional forces where the hanging wall moves downward.
- **Reverse (Thrust) Fault:** Formed under compressional forces where the hanging wall moves upward.
- **Strike-Slip Fault:** Occurs when rocks move horizontally past one another due to shear stress.

1.5 Focus (Hypocenter) and Epicenter



Major Aspect of earthquake

1. **Fault** – A crack in the Earth's crust where rocks are stuck together.
2. **Focus (Hypocenter)** – The point underground where the rocks suddenly break and the earthquake starts.
3. **Epicenter** – The point on the Earth's surface directly above the focus.

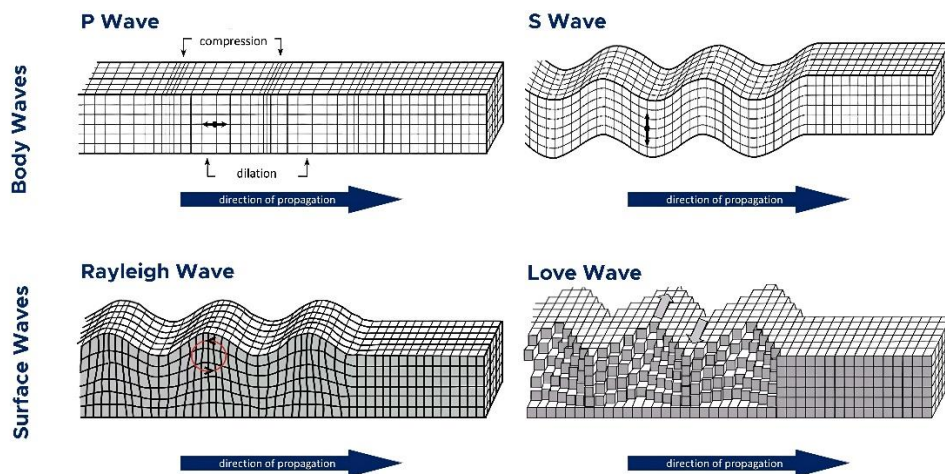
4. **Seismic Waves** – Energy waves that spread outward from the focus, causing the ground to shake.

1.6 Seismic Waves

Seismic waves are vibrations or energy waves produced when an earthquake occurs. They originate from the focus (hypocenter) of the earthquake and travel through the Earth in all directions. Scientists study these waves to determine the location, depth, and magnitude of earthquakes.

How They Are Produced

1. Rocks break or slip along a fault.
2. Energy is released suddenly.
3. The energy travels outward as seismic waves.
4. These waves make the ground move and shake.

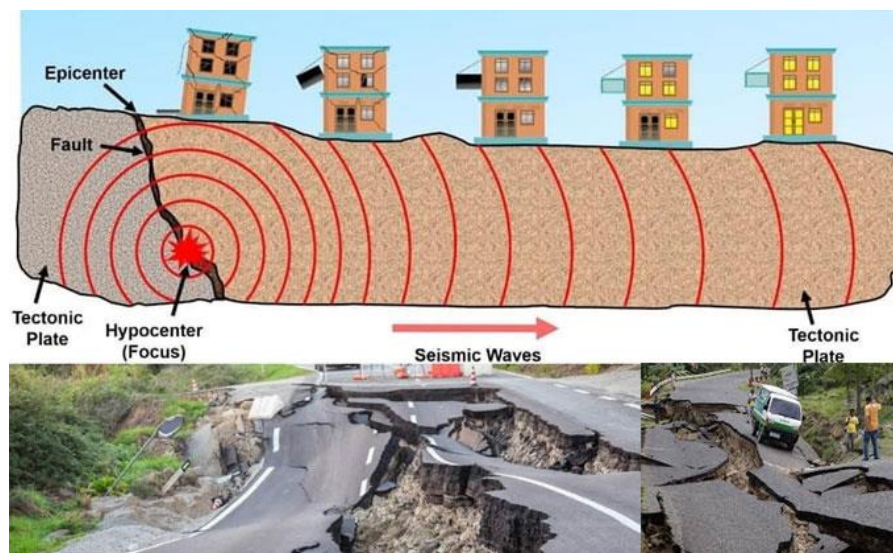


The diagram illustrates the four main types of seismic waves generated during an earthquake. **P-waves (Primary waves)** and **S-waves (Secondary waves)** are **body waves** that travel through the Earth's interior. P-waves are the fastest seismic waves and move through materials by compression and expansion, allowing them to travel through solids, liquids, and gases. S-waves are slower and move particles perpendicular to the direction of travel, so they can pass only through solids. **Love waves** and **Rayleigh waves** are **surface waves** that travel along the Earth's surface. Love waves produce strong horizontal side-to-side motion, while Rayleigh waves create a rolling, ocean-wave-

like movement. Surface waves are generally slower than body waves but cause the greatest damage to buildings and infrastructure because they generate intense ground shaking.

	Wave Type	Subtype	Travels Through	Speed	Damage
1.	Body Wave	P-Wave	Solids, liquids, gases	Fastest	Low
2.	Body Wave	S-Wave	Solids only	Slower	Moderate
3.	Surface Wave	Love Wave	Surface	Slow	High
4.	Surface Wave	Rayleigh Wave	Surface	Slowest	Very High

There are different type of waves that can cause the destruction:



Earthquake Focus



P-waves S-waves



Surface waves spread along the ground

1.7 Measurement of Earthquakes

Earthquakes are measured in terms of magnitude and intensity.

Magnitude

Magnitude represents the total amount of energy released at the earthquake source. It is measured using the Moment Magnitude Scale (Mw), which has largely replaced the Richter Scale for scientific purposes.

Intensity

Intensity measures the effects of an earthquake at a specific location, including damage to buildings and people's experiences.

Richter Scale

Developed by Charles F. Richter in 1935, the Richter Scale measures earthquake magnitude using the amplitude of seismic waves. It is logarithmic, meaning each whole-number increase represents ten times greater wave amplitude.

Moment Magnitude Scale (Mw)

The Moment Magnitude Scale is the modern standard used worldwide. It is based on the seismic moment, considering fault area, amount of slip, and rock strength, making it more accurate for large earthquakes.

Modified Mercalli Intensity (MMI)

The Modified Mercalli Intensity Scale measures earthquake intensity from I (Not Felt) to XII (Total Destruction) based on observed ground shaking and damage.

1.8 Earthquake Monitoring Instruments

Earthquake monitoring relies on specialized instruments to detect and record ground motion.

Seismograph

A seismograph records seismic waves produced during an earthquake and generates a permanent record known as a seismogram.



Source: Field photograph captured during the visit to the India Meteorological Department (IMD), Chandigarh.

Seismometer

A seismometer is a highly sensitive instrument that measures ground motion and provides precise data for earthquake analysis.

Accelerograph

An accelerograph measures strong ground acceleration during earthquakes. It is widely used in engineering studies and for designing earthquake-resistant structures.

1.9 Effects of Earthquakes

Earthquakes can produce both primary and secondary effects.

Primary Effects

- Ground shaking
- Surface faulting
- Structural damage
- Collapse of buildings and infrastructure

Secondary Effects

- Landslides
- Liquefaction
- Fires
- Flooding due to dam or embankment failure
- Tsunamis (in coastal regions)
- Disruption of transportation, communication, and essential services

The severity of these effects depends on the magnitude, depth, geological conditions, construction quality, and preparedness of the affected area.

Chapter 2: Plate Tectonics and Global Seismic Setting

2.1 Introduction to Plate Tectonic Theory

Plate Tectonic Theory is the most widely accepted explanation for the movement of the Earth's lithosphere and the occurrence of earthquakes, volcanic eruptions, and mountain building. Proposed during the 1960s, the theory states that the Earth's outer rigid shell, known as the lithosphere, is broken into several large and small tectonic plates. These plates float over the semi-fluid asthenosphere and move continuously due to convection currents within the Earth's mantle.

The movement of tectonic plates is extremely slow, generally ranging from 2 to 10 centimetres per year. However, when plates interact at their boundaries, stress accumulates in rocks. The sudden release of this stress results in earthquakes. Plate tectonic theory therefore provides the scientific basis for understanding the global distribution of earthquakes and other geological hazards.

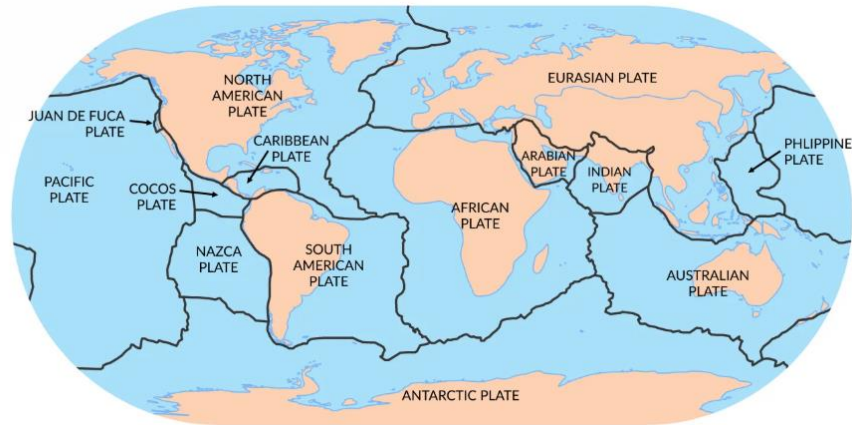
2.2 Major Tectonic Plates of the World

The Earth's lithosphere is divided into seven major plates and several smaller (minor) plates. Their interactions control the occurrence of earthquakes and volcanic activity worldwide.

Major Plates

The seven major tectonic plates are:

- Pacific Plate
- North American Plate
- South American Plate
- Eurasian Plate
- African Plate
- Antarctic Plate
- Indo-Australian Plate (often divided into the Indian Plate and Australian Plate)



These plates cover most of the Earth's surface and move continuously due to mantle convection. Their movement creates plate boundaries where most earthquakes occur.

Minor Plates

In addition to the major plates, several minor plates also contribute to global tectonic activity. Important minor plates include:

- Arabian Plate
- Caribbean Plate
- Nazca Plate
- Philippine Sea Plate
- Cocos Plate
- Scotia Plate
- Juan de Fuca Plate
- Somali Plate
- Anatolian Plate

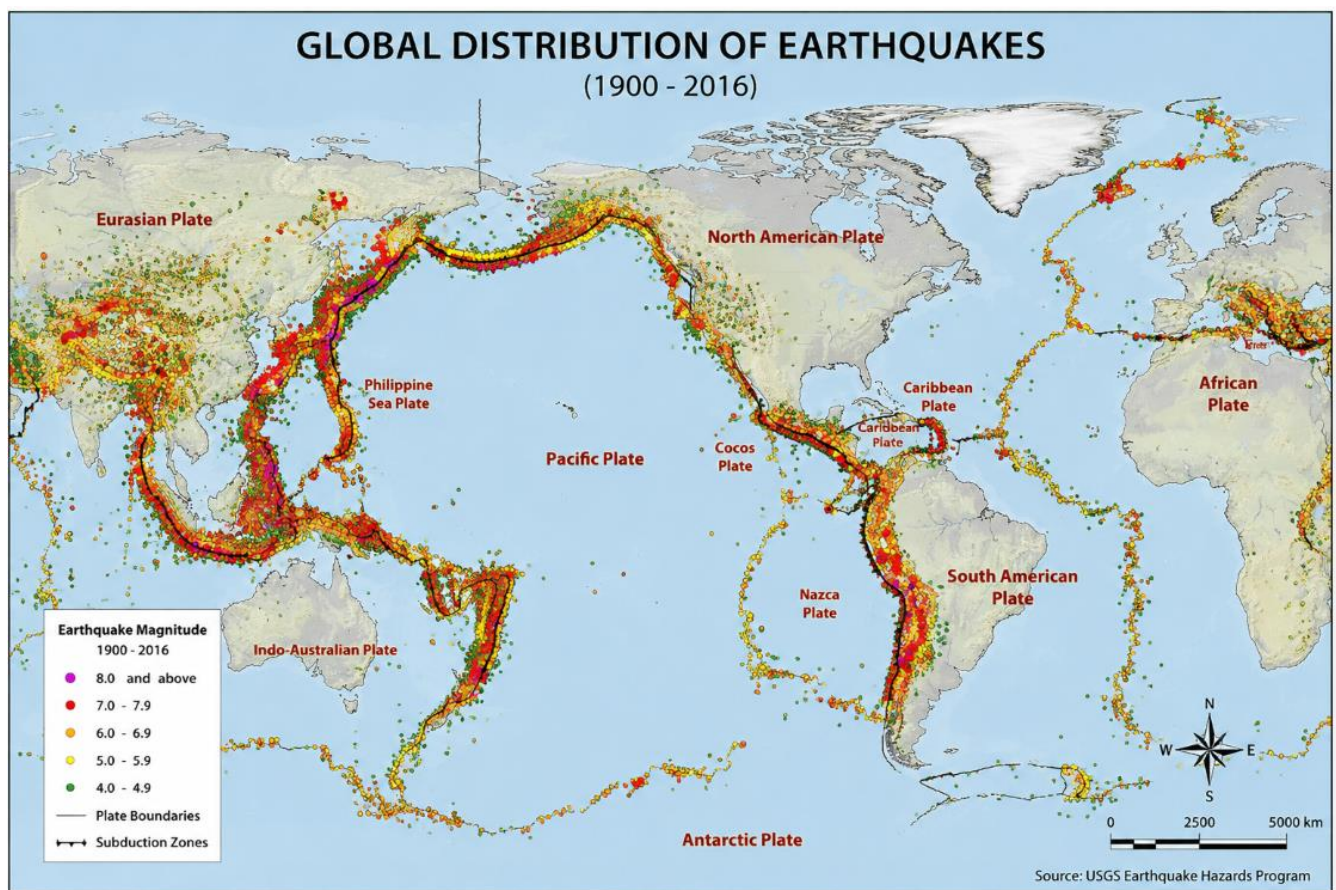
Although smaller in size, these plates are associated with significant seismic and volcanic activity due to their interactions with surrounding major plates.

2.3 Global Distribution of Earthquakes

Earthquakes are not evenly distributed across the Earth. Instead, they are concentrated along active tectonic plate boundaries. The major global earthquake belts include:

- **Pacific Ring of Fire:** The most seismically active region, surrounding the Pacific Ocean and accounting for nearly 75% of the world's earthquakes.
- **Alpine-Himalayan Belt:** Extends from the Mediterranean region through the Middle East to the Himalayas and Southeast Asia. It is the second most active seismic belt and includes northern India.
- **Mid-Ocean Ridge System:** A network of underwater mountain ranges where divergent plate movement causes shallow earthquakes.

Intraplate earthquakes also occur within plate interiors but are less frequent. These are usually associated with the reactivation of ancient faults.



2.4 Tectonic Setting of India

India's tectonic setting plays a crucial role in its seismic activity. The country lies on the **Indian Plate**, which is moving northward at a rate of approximately 5 cm per year and colliding with the **Eurasian Plate**.

Indian Plate and Eurasian Plate

The Indian Plate was once part of the ancient supercontinent Gondwanaland. It drifted northward over millions of years before colliding with the Eurasian Plate around 50 million years ago. The collision continues today, resulting in constant crustal deformation and the accumulation of tectonic stress.

This ongoing convergence makes northern India, particularly the Himalayan region, one of the most earthquake-prone areas in the world.

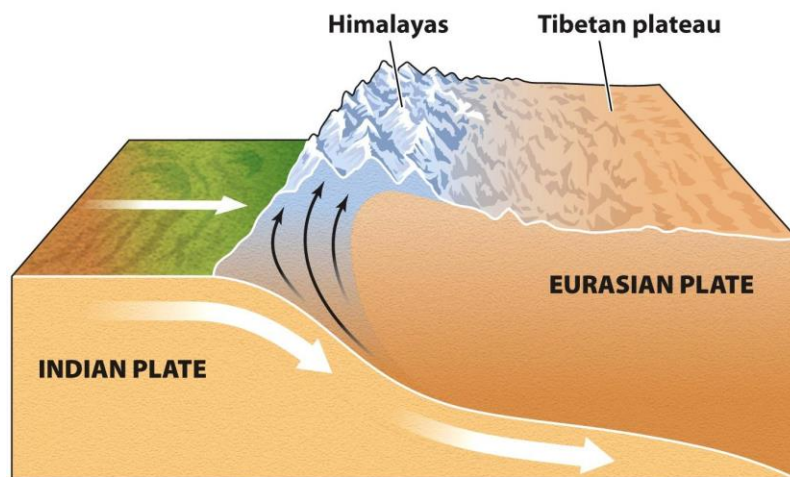


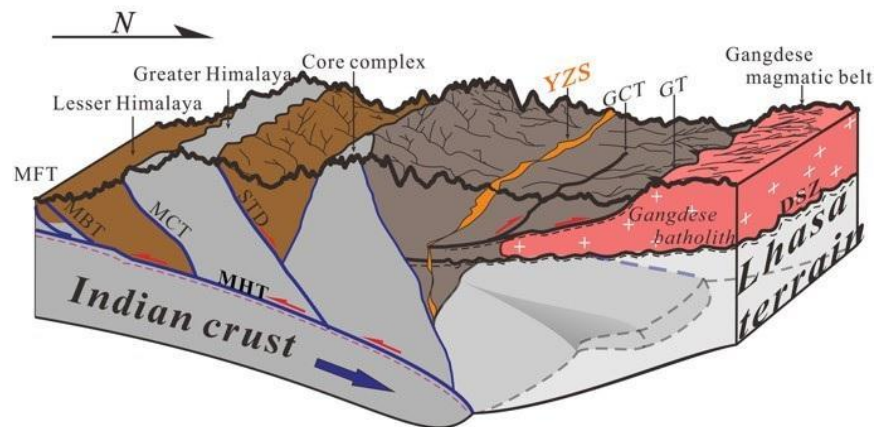
Figure 9-17b
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Formation of the Himalayas

The Himalayan mountain range was formed as a result of the collision between the Indian and Eurasian Plates. Since both plates consist primarily of continental crust, neither plate subducts completely. Instead, the crust is compressed, folded, and uplifted, forming the Himalayas and the Tibetan Plateau.

The collision also produced several major thrust systems, including the Himalayan Frontal Thrust (HFT), Main Boundary Thrust (MBT), and Main Central Thrust (MCT). These active tectonic structures continue to generate earthquakes and significantly influence the seismic hazard of northern India, including Punjab.

Geometric variation of the subducting Indian crust



Explanation:

This diagram shows the Indian Plate moving northward beneath the Eurasian (Tibetan) Plate. The collision compresses and uplifts the Earth's crust, forming the Himalayan Mountains. Major fault zones such as the MFT, MBT, MCT, and MHT develop due to this compression. The Main Himalayan Thrust (MHT) is the primary source of large earthquakes in the Himalayan region. The ongoing convergence of these plates continues to cause mountain building and seismic activity in northern India.

MFT	Main Frontal Thrust
MBT	Main Boundary Thrust
MCT	Main Central Thrust
MHT	Main Himalayan Thrust
STD	South Tibetan Detachment
YZS	Yarlung Zangbo Suture Zone (also called Yarlung Tsangpo Suture Zone)
GCT	Great Counter Thrust
GT	Gangdese Thrust

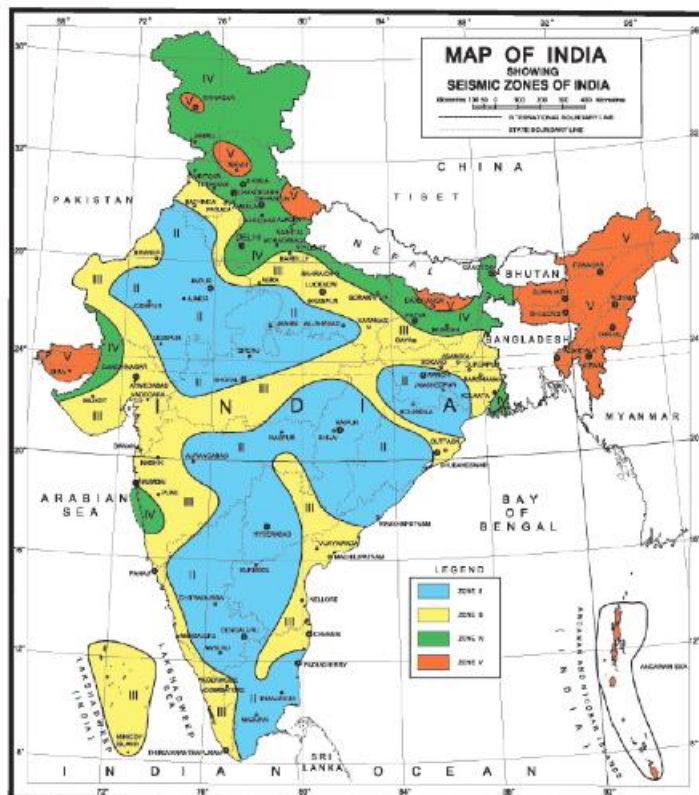
Chapter 3: Seismic Framework of India

3.1 Tectonic Setting of India

India is one of the most earthquake-prone countries in the world due to its unique tectonic setting. The country is situated on the Indian Plate, which is moving northward at a rate of approximately 5 cm per year and colliding with the Eurasian Plate. This collision, which began about 50 million years ago, led to the formation of the Himalayan mountain range and continues to generate immense tectonic stress. The continuous convergence of these plates results in frequent earthquakes, particularly in northern and northeastern India.

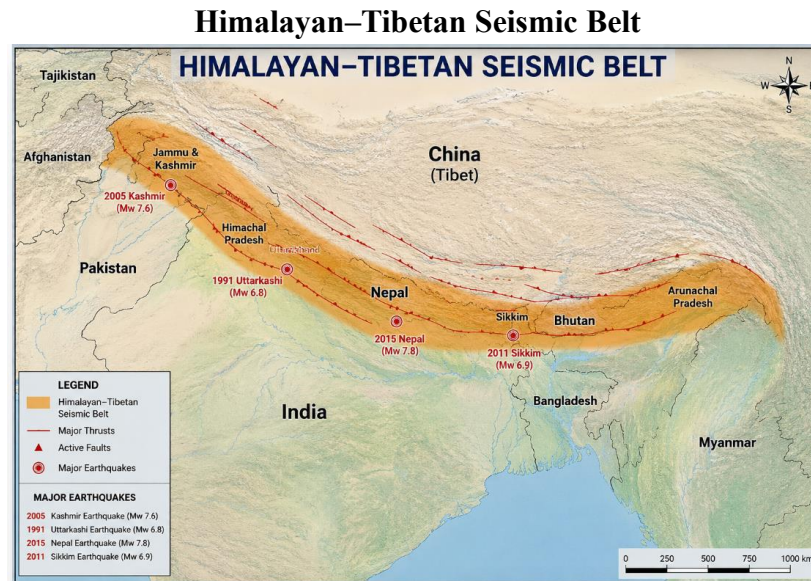
Apart from the Himalayan region, India also experiences intraplate earthquakes within the Peninsular Shield due to the reactivation of ancient geological faults. Significant earthquakes such as the 1993 Latur, 1997 Jabalpur, and 2001 Bhuj earthquakes demonstrate that seismic hazards are not confined only to plate boundaries.

India is divided into seismic zones based on earthquake risk:



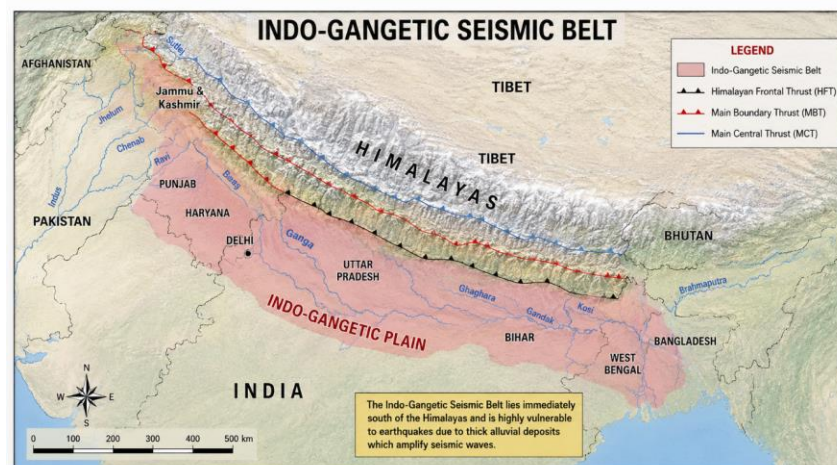
3.2 Himalayan–Tibetan and Indo-Gangetic Seismic Belts

India's seismicity is strongly influenced by the **Himalayan–Tibetan Seismic Belt** and the **Indo-Gangetic Seismic Belt**.



The Himalayan–Tibetan Seismic Belt extends from **Jammu & Kashmir** in the west to **Arunachal Pradesh** in the east. It is one of the world's most active seismic belts due to the ongoing collision between the Indian and Eurasian Plates. This region experiences frequent moderate to great earthquakes and contains several active fault and thrust systems.

Indo-Gangetic Seismic Belt



The Indo-Gangetic Seismic Belt lies immediately south of the Himalayas and includes the Indo-Gangetic Plain. Although this region is relatively stable compared to the Himalayas, it is highly vulnerable to earthquake shaking because of its thick alluvial deposits, which can amplify seismic

waves. States such as **Punjab, Haryana, Delhi, Uttar Pradesh, Bihar, and West Bengal** are influenced by seismic activity originating in the Himalayan region.

3.3 Seismic Zonation of India

To assess earthquake hazards, the **Bureau of Indian Standards (BIS)** has divided India into four seismic zones under **IS 1893 (Part 1)**.

Approximately **59% of India's land area** is vulnerable to moderate or severe earthquakes. Zone V includes parts of the Northeast, Jammu & Kashmir, Himachal Pradesh, Uttarakhand, northern Bihar, and the Andaman & Nicobar Islands. Punjab falls within **Zones II, III, and IV**, while **Chandigarh** lies in **Zone IV**.

Zone	Risk Level
Zone II	Low damage risk
Zone III	Moderate damage risk
Zone IV	High damage risk
Zone V	Very high damage risk

3.4 Major Faults and Thrust Systems

India contains several active faults and thrust systems that are responsible for generating earthquakes. The most significant are:

- **Himalayan Frontal Thrust (HFT):** The youngest and most active thrust along the Himalayan foothills.
- **Main Boundary Thrust (MBT):** Separates the Lesser Himalayas from the Sub-Himalayan region and is associated with frequent seismic activity.
- **Main Central Thrust (MCT):** A major tectonic boundary between the Greater and Lesser Himalayas.
- **Karakoram Fault:** A major strike-slip fault in the northwestern Himalayas.
- **Kachchh Fault System:** Responsible for several damaging earthquakes, including the **2001 Bhuj Earthquake**.

- **Narmada–Son Lineament:** An important intraplate fault zone extending across central India.

These active geological structures accumulate tectonic stress over time, leading to earthquakes when the stored energy is suddenly released.

3.5 Major Earthquakes in India

Year	Earthquake	Magnitude (Mw)	Epicenter / Location		Major Impact
1897	Great Assam Earthquake	8.1	Shillong	Assam Plateau, India;	Severe destruction in Northeast India; widespread ground deformation and infrastructure damage.
1905	Kangra Earthquake	7.8	Kangra, Pradesh	Himachal	Over 20,000 fatalities; extensive damage to buildings and infrastructure.
1934	Bihar–Nepal Earthquake	8.0	Bihar–Nepal Border		Massive destruction in Bihar and Nepal; thousands of deaths and widespread liquefaction.
1950	Assam–Tibet Earthquake	8.6	Assam–Tibet Border		One of India's largest recorded earthquakes; triggered landslides and floods.
1991	Uttarkashi Earthquake	6.8	Uttarkashi, Uttarakhand		Major damage to villages, roads, and public infrastructure.
1993	Latur Earthquake	6.2	Killari, Maharashtra		Over 9,700 deaths; severe destruction despite occurring in a stable continental region.
1997	Jabalpur Earthquake	6.0	Jabalpur, Pradesh	Madhya	Significant structural damage and several casualties.
1999	Chamoli Earthquake	6.6	Chamoli, Uttarakhand		Widespread damage to buildings, roads, and bridges in the Himalayan region.

Year	Earthquake	Magnitude (Mw)	Epicenter / Location	Major Impact
2001	Bhuj Earthquake	7.7	Bhuj, Gujarat	More than 13,000 deaths; extensive destruction of buildings and infrastructure.
2005	Kashmir Earthquake	7.6	Muzaffarabad (Pakistan-administered Kashmir)	Heavy casualties in Jammu & Kashmir and Pakistan; widespread destruction.

Chapter 4: Earthquake Hazard and Vulnerability of Punjab

4.1 Geological and Tectonic Setting of Punjab

Punjab is located in northwestern India and occupies a strategic position between the stable Peninsular Shield and the tectonically active Himalayan region. Although the state is not situated directly on a major plate boundary, it is significantly influenced by the ongoing convergence of the **Indian Plate** and the **Eurasian Plate**. The continued northward movement of the Indian Plate generates tectonic stress in the Himalayas, which is transmitted to the adjoining Indo-Gangetic Plain, including Punjab.

Geologically, Punjab consists mainly of Quaternary alluvial deposits laid down by the **Satluj, Beas, Ravi, and Ghaggar rivers**. These thick layers of unconsolidated sediments can amplify seismic waves during an earthquake, increasing ground shaking and the potential for structural damage. The northern districts of Punjab, located closer to the Himalayan foothills, experience comparatively higher seismic risk than the southern districts.

4.2 Seismic Zonation of Punjab

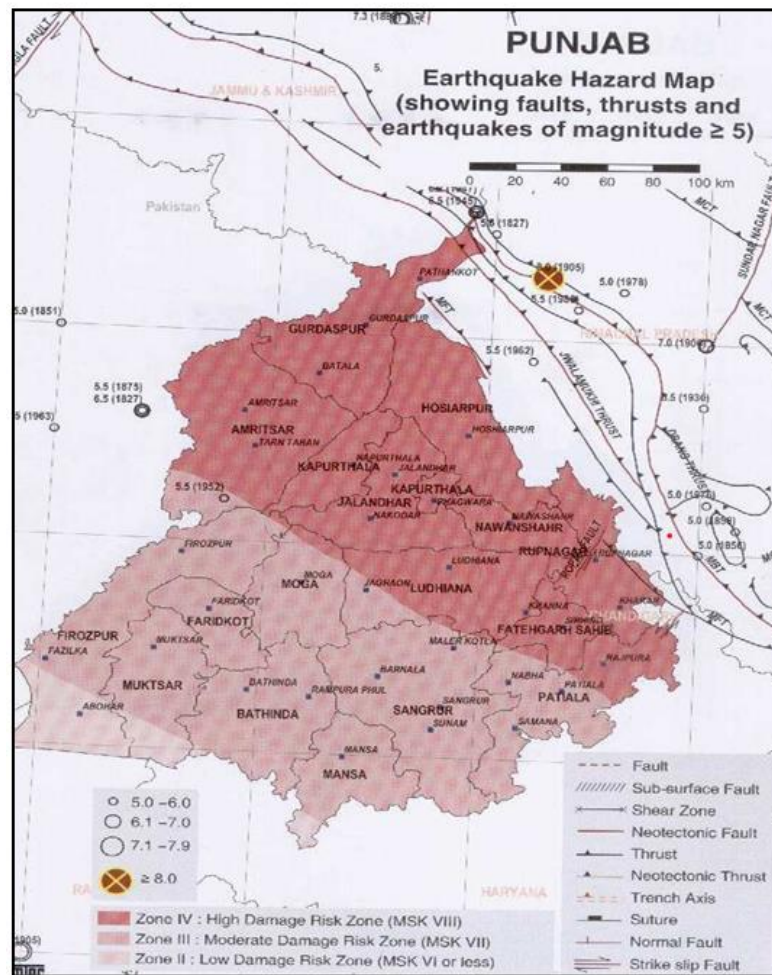
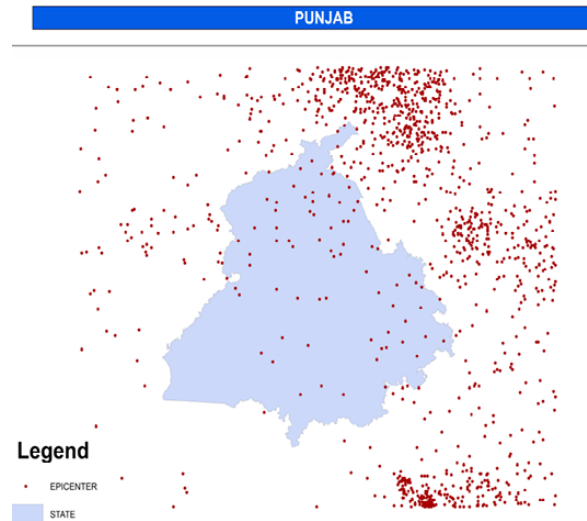
The **Bureau of Indian Standards (BIS)** classifies Punjab into **Seismic Zones II, III, and IV** according to **IS 1893 (Part 1): 2016**.

BIS Seismic Zones in Punjab

Zone II: Low seismic hazard

Zone III: Moderate seismic hazard

Zone IV: High seismic hazard



Source: BMPTC

Northern Punjab falls under Zone IV because of its proximity to the Himalayan Frontal Thrust (HFT), while central districts are mainly in Zone III and parts of southern Punjab fall in Zone II.

Seismic Zone	Districts
Zone IV	Pathankot, Gurdaspur, Hoshiarpur, Rupnagar (Ropar), SBS Nagar (Nawanshahr), SAS Nagar (Mohali), Chandigarh
Zone III	Amritsar, Tarn Taran, Kapurthala, Jalandhar, Ludhiana, Fatehgarh Sahib, Patiala, Sangrur, Barnala, Mansa
Zone II	Bathinda, Faridkot, Muktsar Sahib, Ferozepur, Fazilka

4.3 Active Faults Influencing Punjab

Although Punjab has few active faults within the state itself, its seismicity is strongly influenced by nearby tectonic structures. Major fault systems include:

- Himalayan Frontal Thrust (HFT)
- Main Boundary Thrust (MBT)
- Main Central Thrust (MCT)
- Jammu Fault
- Soan Fault
- Sirhind Fault
- Delhi–Haridwar Ridge
- Kachchh Fault System (indirect influence)

Among these, the **HFT** and **MBT** have the greatest influence on earthquake activity affecting Punjab.

4.4 Earthquake History of Punjab

Punjab has experienced the effects of several major earthquakes originating in the Himalayan region.

Major Historical Earthquakes Affecting Punjab

Year Earthquake Magnitude Impact on Punjab

1905 Kangra	7.8	Severe shaking across northern Punjab
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Year Earthquake Magnitude Impact on Punjab

1934 Bihar–Nepal	8.0	Felt in eastern Punjab
1975 Kinnaur	6.8	Moderate shaking
1991 Uttarkashi	6.8	Felt across Punjab
1999 Chamoli	6.6	Moderate shaking
2001 Bhuj	7.7	Tremors felt throughout Punjab
2005 Kashmir	7.6	Strong shaking in northern districts
2015 Nepal	7.8	Light to moderate tremors across Punjab

4.5 Earthquake Hazard Assessment in Punjab

The earthquake hazard in Punjab depends on several geological and environmental factors:

- Proximity to the Himalayan seismic belt
- Active regional faults
- Thick alluvial soil deposits
- Population density
- Urban expansion
- Critical infrastructure

Primary Hazards

- Ground shaking
- Surface rupture
- Structural collapse

Secondary Hazards

- Soil liquefaction
- Ground settlement
- Fire outbreaks
- Failure of bridges and lifeline infrastructure

- Disruption of transportation and communication
- Damage to dams and canals

4.6 Vulnerability Assessment

Vulnerability refers to the degree to which people, buildings, and infrastructure are susceptible to earthquake damage.

Structural Vulnerability

- Non-engineered masonry buildings
- Old residential structures
- Schools and hospitals
- Bridges and flyovers
- Industrial buildings

Non-Structural Vulnerability

- High population density
- Poor emergency preparedness
- Lack of public awareness
- Failure of electricity and communication systems
- Economic disruption
- Damage to water supply and healthcare services

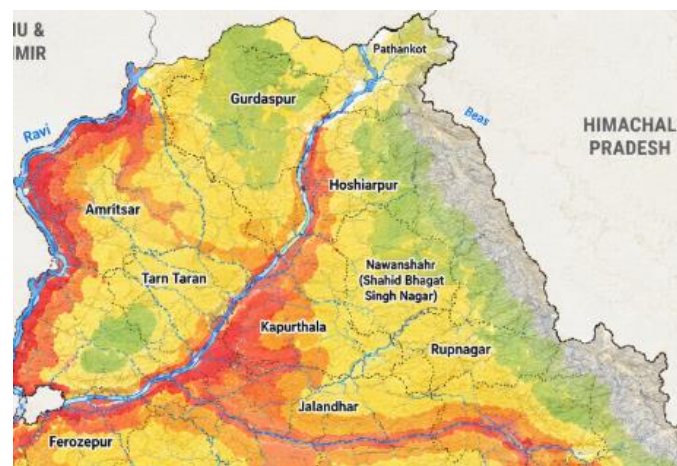
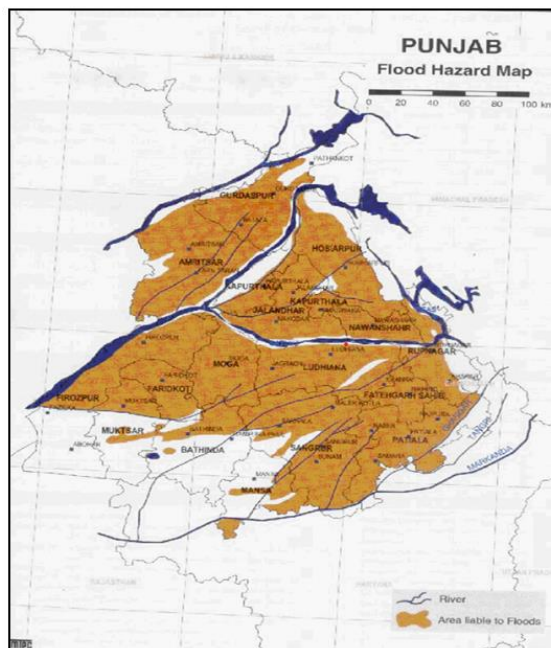
4.7 Liquefaction and Soil Conditions in Punjab

Punjab is dominated by unconsolidated alluvial soils deposited by major river systems. Areas with **soft sediments, high groundwater levels, and poor drainage** are susceptible to **liquefaction** during strong earthquakes.

Liquefaction reduces soil strength and may cause:

- Tilting of buildings
- Settlement of foundations
- Road and bridge damage
- Bursting of underground pipelines

Districts along the **Satluj, Beas, Ravi, and Ghaggar river plains** are relatively more vulnerable to this phenomenon.



Source: Flood Map taken from SDMP

4.8 Risk Assessment of Major Cities

The earthquake risk varies across Punjab depending on seismic zone, population density, construction practices, and soil conditions.

City	Seismic Zone	Risk Level
Chandigarh	IV	High
Mohali	IV	High
Pathankot	IV	High
Gurdaspur	IV	High
Hoshiarpur	IV	High
Amritsar	III	Moderate
Jalandhar	III	Moderate
Ludhiana	III	Moderate
Patiala	III	Moderate
Bathinda	II	Low to Moderate

4.9 Challenges in Earthquake Risk Reduction in Punjab

Punjab faces several challenges in reducing earthquake risk:

- Rapid urbanization
- Non-compliance with BIS earthquake-resistant building codes
- Aging infrastructure
- Lack of retrofitting of vulnerable buildings
- Limited public awareness
- Inadequate emergency preparedness and mock drills
- Insufficient seismic monitoring within the state
- Population growth in urban centres
- Limited integration of GIS-based hazard mapping in planning

Strengthening enforcement of building regulations, improving public awareness, enhancing emergency response capacity, and promoting earthquake-resistant construction are essential for reducing earthquake risk and improving resilience in Punjab.

Chapter 5: Mitigation and Resilience Strategies

Earthquake mitigation and resilience strategies aim to reduce the loss of life, property, and infrastructure by minimizing the impact of earthquakes before they occur. Since earthquakes cannot be prevented, emphasis is placed on preparedness, resilient infrastructure, land-use planning, and community participation. In Punjab, where rapid urbanization and increasing infrastructure development are taking place, adopting effective mitigation measures is essential to reduce future earthquake risks.

5.1 Earthquake-Resistant Construction

Earthquake-resistant construction refers to the design and construction of buildings that can withstand seismic forces without collapsing. The primary objective is to protect human life while minimizing structural damage during an earthquake.

Key principles of earthquake-resistant construction include:

- Designing buildings with regular and symmetrical layouts.
- Providing strong foundations suitable for local soil conditions.
- Using high-quality construction materials such as reinforced concrete and structural steel.
- Ensuring proper reinforcement of beams, columns, and joints.
- Maintaining continuity of the load path from roof to foundation.
- Avoiding soft-story buildings and excessive overhangs.
- Constructing buildings in accordance with seismic design standards.

Modern earthquake-resistant buildings are designed to absorb and dissipate seismic energy through ductile structural elements, reducing the likelihood of collapse.

5.2 Retrofitting of Existing Buildings

Many buildings in Punjab were constructed before modern seismic design standards were introduced. These structures are more vulnerable to earthquake damage and require retrofitting to improve their seismic performance.

Retrofitting involves strengthening existing buildings without demolishing them. Common retrofitting techniques include:

- Jacketing of columns and beams.
- Steel bracing systems.
- Shear wall construction.
- Fiber Reinforced Polymer (FRP) wrapping.
- Foundation strengthening.
- Roof and floor diaphragm strengthening.

Priority should be given to critical facilities such as:

- Hospitals
- Schools and colleges
- Government offices
- Emergency operation center
- Bridges and flyovers
- Lifeline infrastructure

Retrofitting significantly improves the safety and serviceability of buildings during earthquakes.

5.3 BIS Building Codes and Guidelines

The **Bureau of Indian Standards (BIS)** has developed a series of codes to ensure earthquake-resistant design and construction in India. Compliance with these codes is essential for reducing seismic risk.

Important BIS standards include:

BIS Code	Description
IS 1893 (Part 1): 2016	Criteria for Earthquake Resistant Design of Structures
IS 4326: 2013	Earthquake Resistant Design and Construction of Buildings
IS 13828: 1993	Improving Earthquake Resistance of Low-Strength Masonry Buildings
IS 13920: 2016	Ductile Detailing of Reinforced Concrete Structures
IS 13827: 1993	Guidelines for Earthen Buildings

These standards provide recommendations regarding structural design, material quality, reinforcement detailing, foundation design, and construction practices to improve earthquake resilience.

5.4 Land Use Planning and GIS-Based Risk Mapping

Proper land-use planning plays an important role in reducing earthquake risk. Development should avoid highly hazardous areas such as active fault zones, unstable slopes, and areas prone to soil liquefaction.

Geographic Information System (GIS) technology is widely used for earthquake hazard assessment and planning. GIS helps in:

- Preparing seismic hazard maps.
- Identifying vulnerable populations and infrastructure.
- Mapping liquefaction-prone areas.
- Assessing exposure of critical facilities.
- Supporting emergency response planning.
- Assisting urban planners in selecting safer locations for future development.

In Punjab, GIS-based hazard mapping can help identify districts and urban center with higher seismic vulnerability, enabling better planning and resource allocation.

5.5 Community-Based Disaster Risk Reduction

Community participation is a key component of earthquake risk reduction. A well-informed and prepared community can significantly reduce casualties and improve response during disasters.

Important community-based measures include:

- Public awareness campaigns on earthquake safety.
- Regular earthquake mock drills in schools, colleges, hospitals, and offices.
- Training in first aid and basic search-and-rescue techniques.
- Formation of Community Emergency Response Teams (CERTs).
- Preparation of household emergency plans and emergency kits.
- Strengthening coordination among government agencies, local authorities, NGOs, and volunteers.

- Promoting earthquake insurance and financial preparedness.

The **National Disaster Management Authority (NDMA)**, **Punjab State Disaster Management Authority (PSDMA)**, and district administrations play an important role in organizing awareness programs and strengthening disaster preparedness across the state.

Community resilience can be further enhanced through continuous education, early warning dissemination (where applicable), and integration of disaster risk reduction into school curricula and local development planning.

Chapter 6: Case Studies

Case studies provide valuable insights into the causes, impacts, response mechanisms, and lessons learned from major earthquakes. Although Punjab has not experienced a catastrophic earthquake in recent history, it is vulnerable to strong ground shaking from earthquakes occurring in the Himalayan region and other parts of India. The **1905 Kangra Earthquake** and the **2001 Bhuj Earthquake** are significant examples that highlight the importance of earthquake-resistant infrastructure, disaster preparedness, and effective emergency management. The lessons learned from these events are directly applicable to earthquake risk reduction in Punjab.

6.1 The 1905 Kangra Earthquake

Introduction

The **Kangra Earthquake** occurred on **4 April 1905** in the Kangra Valley of present-day Himachal Pradesh. With a **moment magnitude (M_w) of approximately 7.8**, it is considered one of the most destructive earthquakes in Indian history. The earthquake originated due to movement along the active Himalayan thrust system associated with the ongoing collision between the Indian and Eurasian Plates.

Causes

- Collision of the Indian Plate with the Eurasian Plate.
- Sudden rupture along the Himalayan Frontal Thrust (HFT) and associated fault systems.
- Accumulation and sudden release of tectonic stress.

Impact

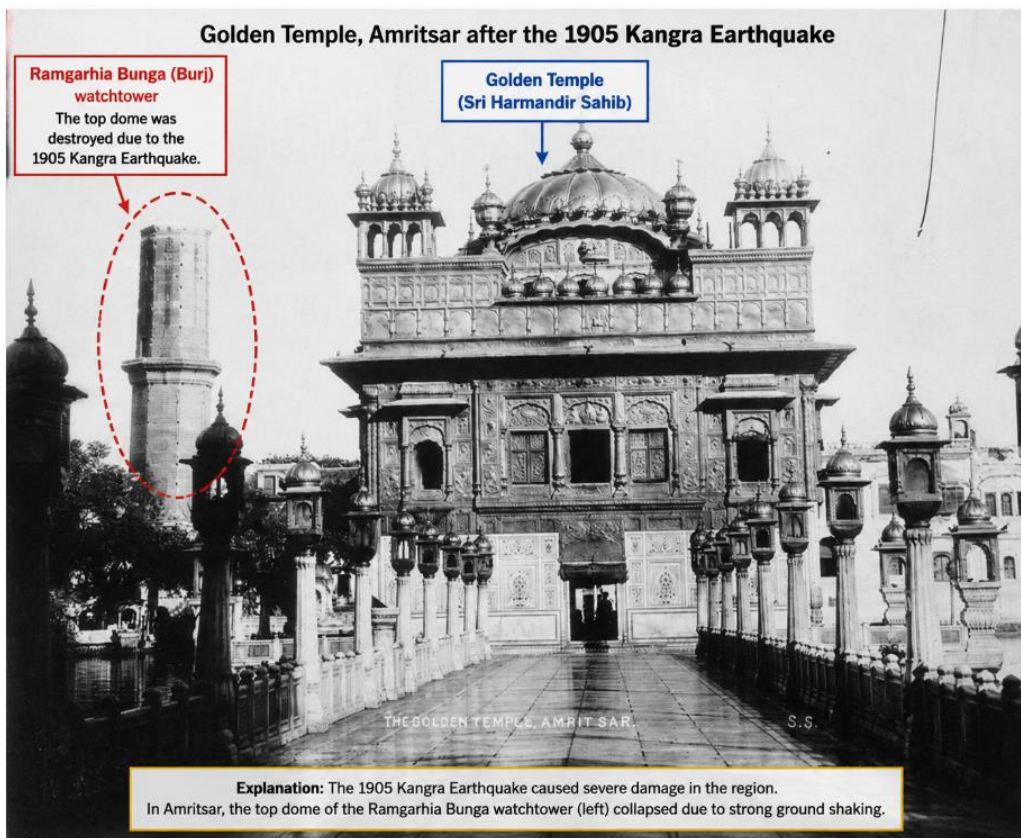
- More than **20,000 people lost their lives**.
- Extensive destruction in Kangra, Dharamshala, Palampur, and surrounding areas.

- Thousands of buildings collapsed, including temples, schools, hospitals, and government buildings.
- Roads, bridges, and communication systems were severely damaged.
- Strong tremors were felt across **Punjab**, causing panic and minor structural damage in several districts.





Photograph of the ruins of Bajreshwari Mata Temple, Kangra [also known as Brajeshwari Devi Temple, Bajreshwari Mata Temple, or Kangra Devi Temple] taken in the aftermath of the 1905 Kangra earthquake. The earthquake occurred on 4 April 1905, meaning this photograph must have been snapped shortly after. Another photograph of the same ruins is **



Photograph of the Golden Temple in Amritsar after the earthquake. The top dome of the Ramgarhia Bunga's Burj watchtower to the left was destroyed due to the earthquake.

Disaster Response

- Rescue operations were carried out by the British administration and local communities.
- Temporary shelters and medical camps were established.
- Relief supplies, food, and medical assistance were provided to affected populations.

Lessons for Punjab

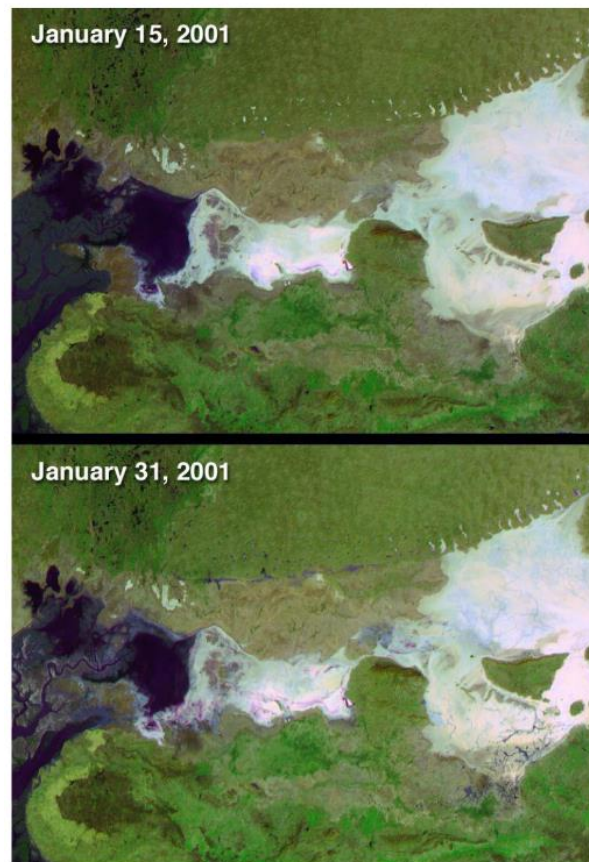
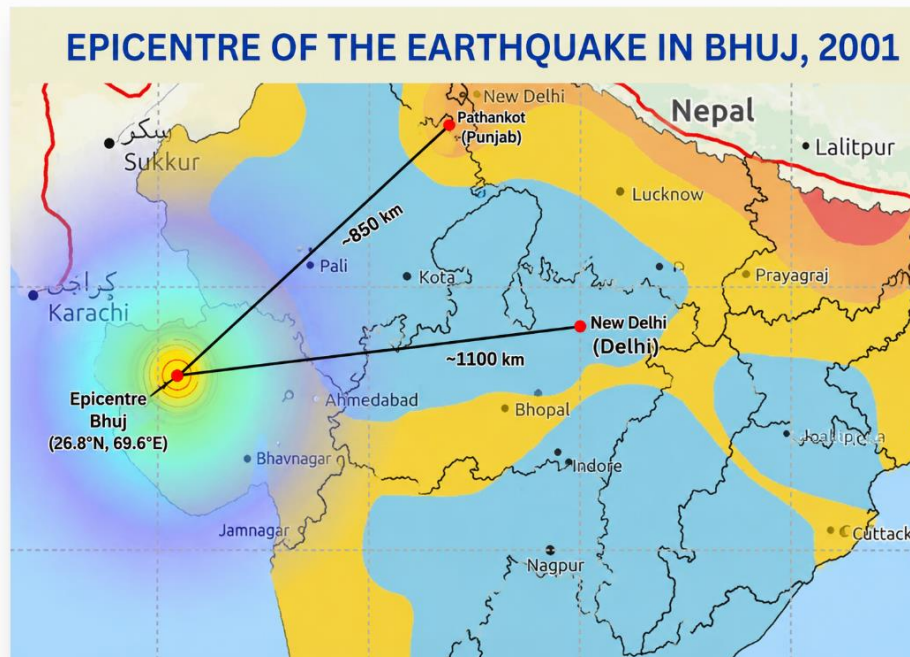
- Strict enforcement of earthquake-resistant building codes.
 - Seismic safety assessment of old masonry buildings.
 - Strengthening hospitals, schools, and public infrastructure.
 - Improved emergency preparedness and public awareness.
 - Enhanced seismic monitoring in northern India.
-

6.2 The 2001 Bhuj Earthquake

Introduction

The **Bhuj Earthquake** occurred on **26 January 2001** in the Kutch district of Gujarat. It had a **moment magnitude (M_w) of 7.7** and remains one of the deadliest earthquakes in modern India. Although it occurred in western India, the earthquake was felt across several states, including Punjab.

Year	Location	Magnitude
1872	Bhavnagar	5.0
1919	Bhavnagar	5.7
1938	Paliyad village	5.0-5.7
1956	Anjar, Kutch	6.1
1969	Mount Abu Region	5.5
1970	Bharuch	5.4
2001	Bhuj	7.6-7.7
2020	Kuch Region	5.8



Comparison of satellite images from 15 January and 31 January 2001 clearly shows major environmental changes following the 26 January 2001 Bhuj earthquake. The post-earthquake image reveals numerous new water-filled areas, dark sediment deposits, and linear ground disturbances caused by liquefaction and groundwater expulsion. These changes indicate that

intense seismic shaking significantly altered the surface of the Rann of Kachchh, with earthquake effects extending far beyond the epicentral region

These images show that earthquakes do not only damage buildings—they also change the landscape. The Bhuj earthquake altered groundwater flow, produced widespread liquefaction, and modified the surface of the Rann of Kachchh. Satellite imagery allowed scientists to detect these changes rapidly, including in remote areas that were difficult to access on the ground.



Causes

- Reactivation of ancient faults within the Kutch Rift Zone.
- Intraplate tectonic stress within the Indian Plate.
- Sudden fault rupture beneath the Kutch region.

Impact

- More than **13,000 people were killed**.
- Around **167,000 people were injured**.
- Nearly **400,000 houses were destroyed**.
- Extensive damage to roads, bridges, hospitals, schools, industries, and public utilities.
- Economic losses exceeded **US\$5 billion**.
- Tremors were experienced in many parts of northern India, including Punjab.

Disaster Response

- Large-scale rescue operations by the Armed Forces, NDRF (later established), NGOs, and volunteers.
- Establishment of temporary hospitals and relief camps.
- Rapid restoration of transportation, communication, and essential services.
- International humanitarian assistance.
- Long-term rehabilitation and reconstruction programmes.

Policy Changes After the Bhuj Earthquake

The Bhuj earthquake led to major reforms in India's disaster management system:

- Establishment of the **National Disaster Management Authority (NDMA)**.
- Enactment of the **Disaster Management Act, 2005**.
- Formation of the **National Disaster Response Force (NDRF)**.
- Revision of **BIS seismic design codes**.
- Greater emphasis on earthquake-resistant construction and retrofitting.
- Expansion of seismic monitoring by the National Centre for Seismology (NCS).

Lessons for Punjab

- Need for district-level earthquake preparedness plans.
- Retrofitting of vulnerable public buildings.
- Adoption of GIS-based hazard mapping.
- Regular mock drills and community awareness programmes.

- Strengthening coordination among disaster management agencies.
 - Enforcement of BIS earthquake-resistant construction standards.
-

Comparative Analysis of the Two Earthquakes

Parameter	1905 Kangra Earthquake	2001 Bhuj Earthquake
Date	4 April 1905	26 January 2001
Magnitude (Mw)	7.8	7.7
Epicenter	Kangra, Himachal Pradesh	Bhuj, Gujarat
Tectonic Setting	Himalayan plate collision	Intraplate faulting (Kutch Rift Zone)
Death Toll	>20,000	>13,000
Major Damage	Buildings, temples, roads	Houses, industries, infrastructure
Effect on Punjab	Strong shaking and minor damage	Tremors felt across the state
Key Lesson	Seismic-safe construction in Himalayan region	Modern disaster management and resilient infrastructure

Conclusion

The **1905 Kangra** and **2001 Bhuj** earthquakes demonstrate that both plate-boundary and intraplate earthquakes can cause catastrophic consequences. They highlight the importance of scientific hazard assessment, enforcement of seismic building codes, preparedness planning, and resilient infrastructure. For Punjab, these case studies reinforce the need to strengthen earthquake risk reduction measures, particularly in northern districts and rapidly urbanizing areas, to minimize future losses. The **1905 Kangra Earthquake** and the **2001 Bhuj Earthquake** demonstrate that earthquakes occurring in different tectonic environments can both result in catastrophic human and economic losses. The Kangra earthquake originated along the active Himalayan plate boundary and significantly affected Punjab due to its close proximity, causing strong ground shaking and damage to historical structures such as the Ramgarhia Bunga at the Golden Temple, Amritsar. In contrast, the Bhuj earthquake occurred within the stable continental region of Gujarat, proving that destructive earthquakes can also occur away from plate boundaries due to the reactivation of ancient faults. Both earthquakes exposed the vulnerability of poorly designed buildings, inadequate preparedness, and weak disaster response systems. Although Punjab was not the epicentral region in either event, these earthquakes highlight that the state remains vulnerable because of its proximity to the Himalayan Seismic Belt, thick alluvial soils that amplify seismic waves, increasing urbanization, and the presence of numerous non-engineered buildings. With rapid infrastructure development and growing population density, a future Himalayan earthquake of similar magnitude could result in greater damage than experienced in the past unless appropriate mitigation measures are adopted.

Chapter 7: Findings, Recommendations and Conclusion

7.1 Key Findings

The study revealed that although Punjab is not situated directly on an active plate boundary, it remains vulnerable to earthquakes due to its proximity to the **Himalayan Seismic Belt**, one of the most active seismic regions in the world. According to the **Bureau of Indian Standards (BIS)** seismic zonation map, Punjab falls within **Seismic Zones II, III, and IV**, with the northern districts facing comparatively higher seismic risk.

The geological characteristics of Punjab, particularly its thick alluvial deposits and high groundwater table, can amplify seismic waves and increase the likelihood of **soil liquefaction** during strong earthquakes. Historical events such as the **1905 Kangra Earthquake** and the **2001 Bhuj Earthquake** demonstrate that even earthquakes occurring outside Punjab can produce noticeable ground shaking and damage, highlighting the state's vulnerability.

Rapid urbanization, increasing population density, unplanned development, and the presence of many non-engineered buildings have further increased earthquake risk. While institutions such as the **National Centre for Seismology (NCS)**, **India Meteorological Department (IMD)**, **National Disaster Management Authority (NDMA)**, and **Punjab State Disaster Management Authority (PSDMA)** have strengthened earthquake monitoring and preparedness, continued efforts are needed to improve resilience.

7.2 Recommendations

Based on the findings of this study, the following recommendations are proposed to reduce earthquake risk in Punjab:

- Strictly enforce **BIS earthquake-resistant building codes** in all new construction.
- Conduct **structural safety audits** and retrofit vulnerable schools, hospitals, government buildings, bridges, and heritage structures.
- Prepare and regularly update **district-level GIS-based seismic hazard and liquefaction maps**.
- Strengthen land-use planning by avoiding development in highly vulnerable areas and ensuring site-specific geotechnical investigations before major projects.
- Expand the network of **seismic observatories and strong-motion accelerographs**, particularly in northern Punjab.
- Organize regular **earthquake awareness campaigns and mock drills** in schools, colleges, hospitals, industries, and residential communities.

- Improve coordination among **NDMA, PSDMA, district administrations, IMD, NCS, and emergency response agencies.**
- Develop well-equipped emergency operation centres with trained search-and-rescue teams and adequate medical resources.
- Encourage universities and research institutions to undertake studies on seismic hazard assessment, soil behaviour, and earthquake engineering in Punjab.
- Promote community participation in disaster preparedness through training in first aid, evacuation procedures, and household emergency planning.

7.3 Conclusion

- Earthquakes remain one of the most destructive natural hazards because they occur suddenly and can cause widespread damage to lives, infrastructure, and the economy. Although Punjab is not located directly on a major tectonic plate boundary, its proximity to the Himalayas and its geological conditions make it susceptible to significant earthquake impacts. The lessons learned from the **1905 Kangra Earthquake** and the **2001 Bhuj Earthquake** highlight the importance of earthquake-resistant construction, scientific hazard assessment, effective disaster management, and public preparedness.
 - Building a resilient Punjab requires a coordinated approach involving government agencies, technical institutions, local authorities, and communities. The effective implementation of seismic building codes, retrofitting of vulnerable structures, improved hazard mapping, strengthened monitoring systems, and increased public awareness can substantially reduce future earthquake losses. By integrating disaster risk reduction into development planning and promoting a culture of preparedness, Punjab can enhance its resilience to future seismic events and ensure safer and more sustainable development.
-
- This chapter serves as a comprehensive conclusion to your report by summarizing the major findings, providing practical recommendations, and reinforcing the importance of earthquake risk reduction in Punjab.
-

Chapter 8: References

Include all books, reports, journals, websites, and government documents used in your report. Use one citation style consistently (APA 7th edition is commonly accepted).

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 - Punjab State Disaster Management Authority (PSDMA): <https://psdma.punjab.gov.in>
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Chapter 9: Appendices

Appendices contain supporting material that is useful but too detailed to include in the main chapters.

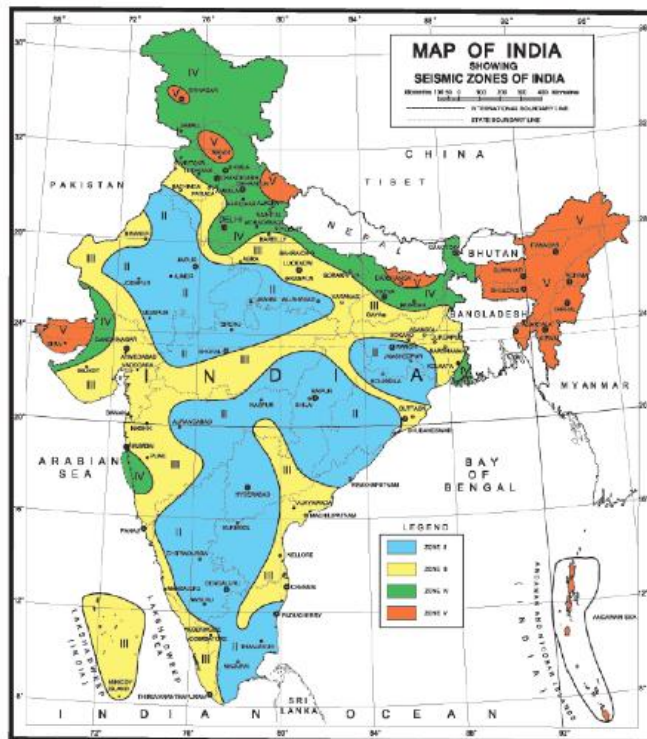
Appendix A: BIS Seismic Zonation Map of India

- Latest BIS Seismic Zone Map (IS 1893:2016)

Appendix B: District-wise Seismic Zonation of Punjab

District	BIS Seismic Zone
Pathankot	IV
Gurdaspur	IV
Hoshiarpur	IV
Rupnagar (Ropar)	IV
SAS Nagar (Mohali)	IV
Chandigarh (UT)	IV
Remaining districts	II/III (as per BIS map)

Appendix C: Major Earthquakes in India



Appendix D: Earthquake Preparedness Guidelines

- Drop, Cover, and Hold On.

- Emergency Kit Checklist.
 - Family Emergency Plan.
 - Evacuation Procedure.
-

Appendix E: Field Visit to IMD, Chandigarh

Include:

- A field visit was conducted to the IMD, Chandigarh, where a seismic monitoring instrument (seismograph) was observed. Although detailed information about its operation was not provided during the visit, it offered an opportunity to observe the equipment used for earthquake monitoring and understand its role in recording seismic activity.
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Appendix D: Earthquake Preparedness Guidelines

- Drop, Cover, and Hold On.
 - Emergency Kit Checklist.
 - Family Emergency Plan.
 - Evacuation Procedure.
-